

WASP-50b

R-1578

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_141130 · created 2026-08-01T23:37:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456991.688 +/- 0.016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.025
Transit depth (Rp/Rs)^2	1.48 +/- 0.61 [%]
Orbital Inclination (inc)	85.82 +/- 1.96
Airmass coefficient 1 (a1)	0.971 +/- 0.011
Airmass coefficient 2 (a2)	0.021 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	#1 - [267, 462]
Optimal Aperture	3.78
Optimal Annulus	4.87
Transit Duration (day)	0.074 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.025 vs 0.1404 (deviation 0.74σ)
Duration fit vs published	0.074 d vs 0.0754 d (deviation 0.09σ)
Mid-time O–C	-11.2 min
Depth SNR	4.9
Residual scatter	1.14 %

Light curve

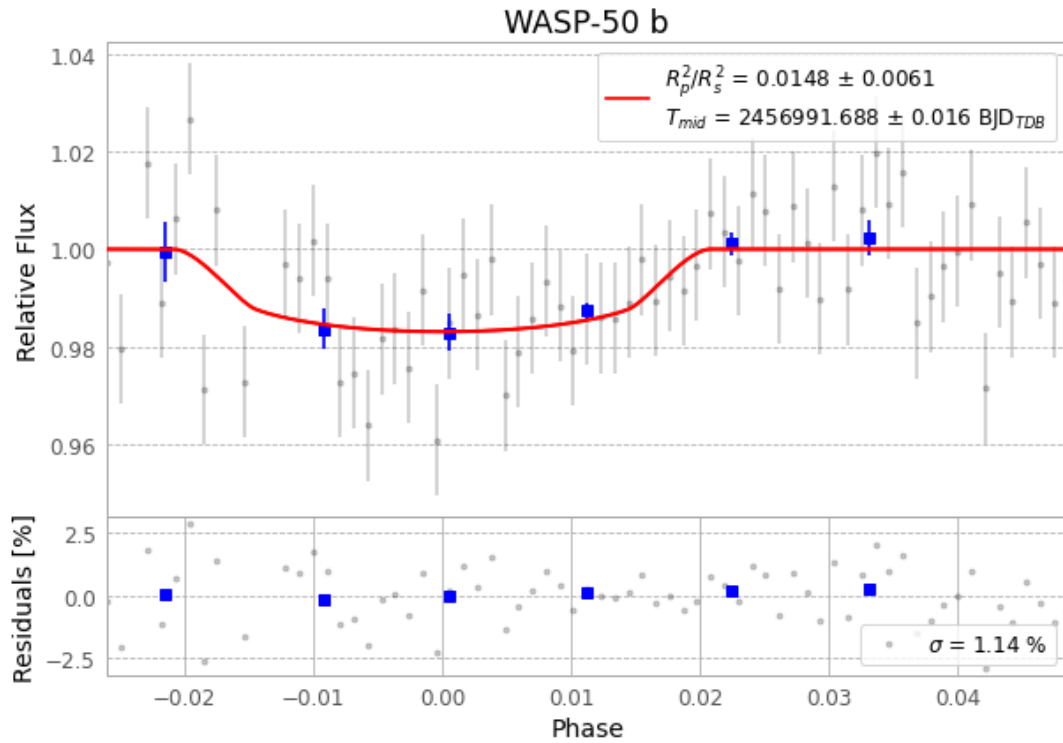


Figure 1 — FinalLightCurve_WASP-50 b_2014-11-29.png (SHA-256 6366e47e440d775f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0485a65ca9c8a215...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), WASP-50141130030912.FITS → WASP-50141130063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-141130.log (9ee97d3a8ed1...), FinalLightCurve_WASP-50 b_2014-11-29.png (6366e47e440d...), FinalLightCurve_WASP-50 b_2014-11-29.csv (7e4853682755...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 23:37Z.